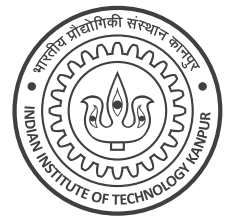


CS 771A: Intro to Machine Learning, IIT Kanpur			Quiz II (02 Apr 2026)	
Name				20 marks
Roll No		Dept.		Page 1 of 2

Instructions:

1. This question paper contains 1 page (2 sides of paper). Please verify.
2. Write your name, roll number, department above in **block letters neatly with ink**.
3. Write your final answers neatly **with a blue/black pen**. Pencil marks may get smudged.
4. Don't overwrite/scratch answers especially in MCQ – such cases may get straight 0 marks.
5. Do not rush to fill in answers. You have enough time to solve this quiz.



Q1 (Machine Unlearning) Melbo learnt a model $\mathbf{w} \in \mathbb{R}^d$ (hidden bias) via least squares regression on N points $X \in \mathbb{R}^{N \times d}, \mathbf{y} \in \mathbb{R}^N$ with $\mathbf{x}^i \in \mathbb{R}^d, y^i \in \mathbb{R}, i \in [N]$. Derive an expression for $\mathbf{w}$ assuming $X^T X$ is invertible. Melba filed a DPDP complaint asking the j^{th} data point (containing Melba's personal data) to be removed from model training. Melbo, being lazy, did not remove $\mathbf{x}^j$ from the dataset and instead, set $y^j = 0$ and learnt a new model $\tilde{\mathbf{w}}$ using least squares on those same N points. Derive an expression for $\mathbf{w} - \tilde{\mathbf{w}}$. Simplify it as much as possible. **(2 + 4 = 6 marks)**

Q2 (IPL Intrigue). Melbo is a big IPL fan and is trying to analyse the performance of KKR vs SRH on various kinds of pitches. Let K be the event that KKR won a KKR-vs-SRH match and S be the event that SRH won a KKR-vs-SRH match. There are 3 kinds of pitches $F = \text{flat}, G = \text{green}, D = \text{dusty}$. A total of 32 KKR-vs-SRH matches were played, 37.5% of which were on dusty pitches. Equal number of matches were played on flat and green pitches. The two teams are equally matched i.e. $\mathbb{P}[K] = \mathbb{P}[S]$. On flat pitches, KKR has a $4 \times$ chance of winning i.e. $\mathbb{P}[K | F] = 4 \cdot \mathbb{P}[S | F]$. This advantage flips exactly on green pitches i.e. $\mathbb{P}[K | G] = \mathbb{P}[S | G]$. Find out the following quantities as fractions or decimals (no derivations needed). *Hint: either use Bayes rule or fill-up a 2×3 matrix showing which team won how many matches on what kind of pitch.* **(8 x 1 = 8 marks)**

$$\mathbb{P}[K \mid D] =$$

$$\mathbb{P}[S \mid D] =$$

$$\mathbb{P}[F \mid K] =$$

$$\mathbb{P}[F \mid S] =$$

$$\mathbb{P}[G \mid K] =$$

$$\mathbb{P}[G \mid S] =$$

$$\mathbb{P}[D \mid K] =$$

$$\mathbb{P}[D \mid S] =$$

Q3 (Gaussian Regularization?) Consider a data point $\mathbf{x} \in \mathbb{R}^d, y \in \mathbb{R}$, linear model $\mathbf{w} \in \mathbb{R}^d$ (hidden bias), a noise vector $\mathbf{z} \in \mathbb{R}^d \sim \mathcal{N}(\mathbf{0}, I_d)$ sampled from a standard normal distribution, and a noisy version of the least squares loss $\ell(\mathbf{w}, \mathbf{z}) \stackrel{\text{def}}{=} \frac{1}{2} \left(y - \mathbf{w}^\top (\mathbf{x} + \mathbf{z}) \right)^2$. Write down an expression for the gradient of the noisy least squares loss i.e., $\nabla_{\mathbf{w}} \ell(\mathbf{w}, \mathbf{z})$ (no derivation needed). Then, derive the expected gradient $\mathbb{E}_{\mathbf{z}}[\nabla_{\mathbf{w}} \ell(\mathbf{w}, \mathbf{z})]$. *Note: the gradient is w.r.t. the model $\mathbf{w}$ while the expectation is w.r.t. the noise term $\mathbf{z}$.* Simplify your expressions as much as possible. Note that $\mathbb{E}_{\mathbf{z}}[\nabla_{\mathbf{w}} \ell(\mathbf{w}, \mathbf{z})]$ shouldn't contain any $\mathbf{z}$ terms although $\nabla_{\mathbf{w}} \ell(\mathbf{w}, \mathbf{z})$ will contain $\mathbf{z}$ terms. **(2 + 4 = 6 marks)**

$$\nabla_{\mathbf{w}} \ell(\mathbf{w}, \mathbf{z}) =$$

Write final expression in the box

$$\mathbb{E}_{\mathbf{z}}[\nabla_{\mathbf{w}} \ell(\mathbf{w}, \mathbf{z})] =$$

Give brief derivation with simplification below